

House Price Prediction

A Machine Learning Approach to Real Estate Valuation

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The Challenge in Real Estate Valuation

The real estate industry continues to depend heavily on traditional manual appraisals, a process fraught with limitations that impact buyers, sellers, and investors alike.



Subjective Pricing

Different agents provide varying price estimates based on personal bias and experience, leading to inconsistent valuations for similar properties.



Inconsistent Results

High risk of human error in calculations and assessments undermines confidence in property valuations and market transparency.



Time-Intensive Process

Physical site visits, extensive paperwork, and lengthy approval cycles create delays that frustrate stakeholders and slow down transactions.

The market needed a faster, more objective, and scientifically grounded approach to property valuation.



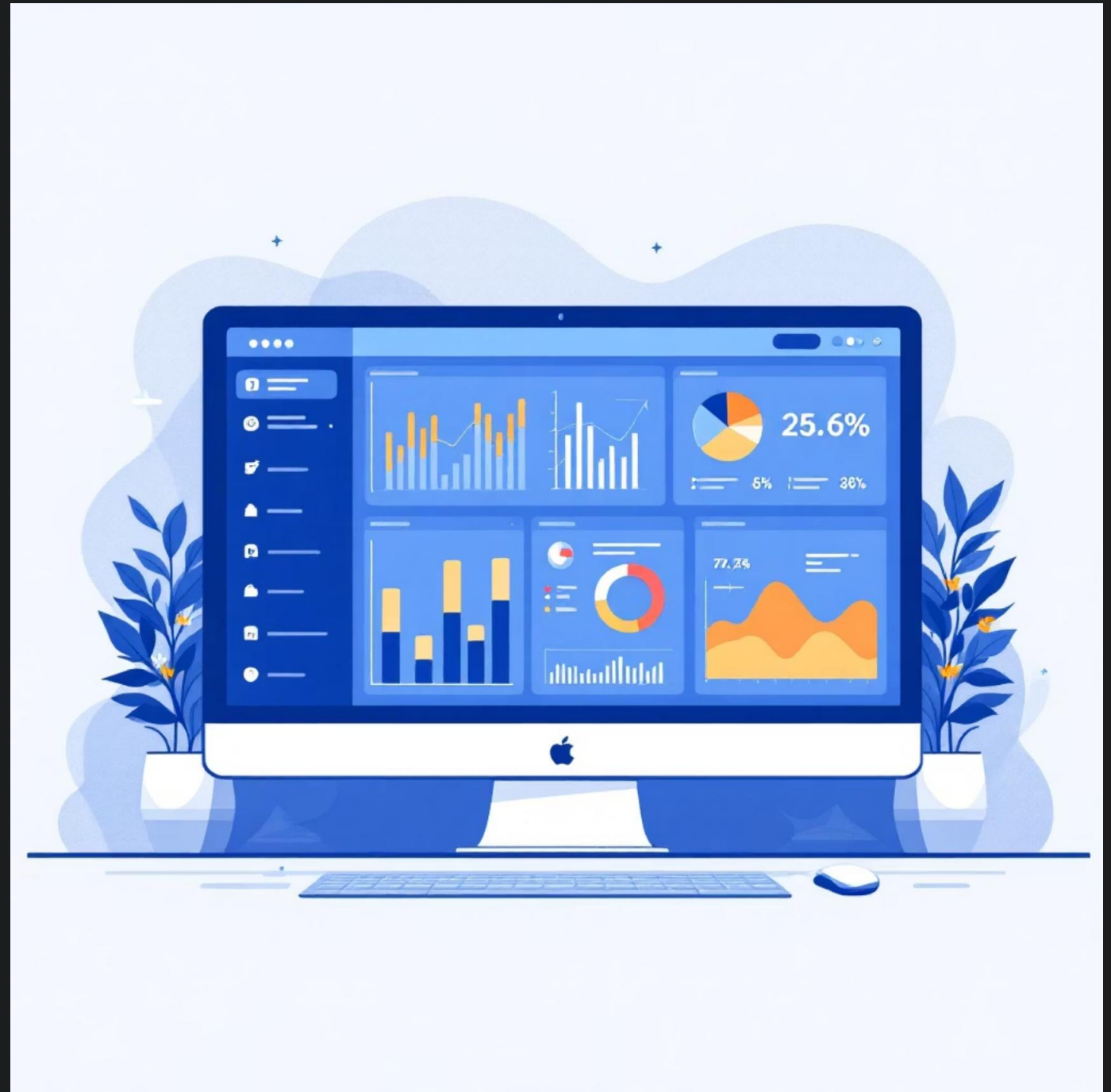
Our Solution: Intelligent Automation

Data-Driven Predictions

Our web application leverages Random Forest Regression to analyse historical housing data and deliver instant, accurate price estimates.

The system eliminates guesswork by processing key property attributes—median income, house age, location coordinates, and more—to generate scientific valuations.

With automatic currency conversion between USD and INR, the platform serves diverse users across markets seamlessly.



System Architecture

Our solution is built on a robust three-tier architecture that ensures scalability, maintainability, and optimal performance.

01

Frontend Layer

Built with HTML, CSS, and JavaScript to create an intuitive user interface that collects property details and displays results elegantly.

02

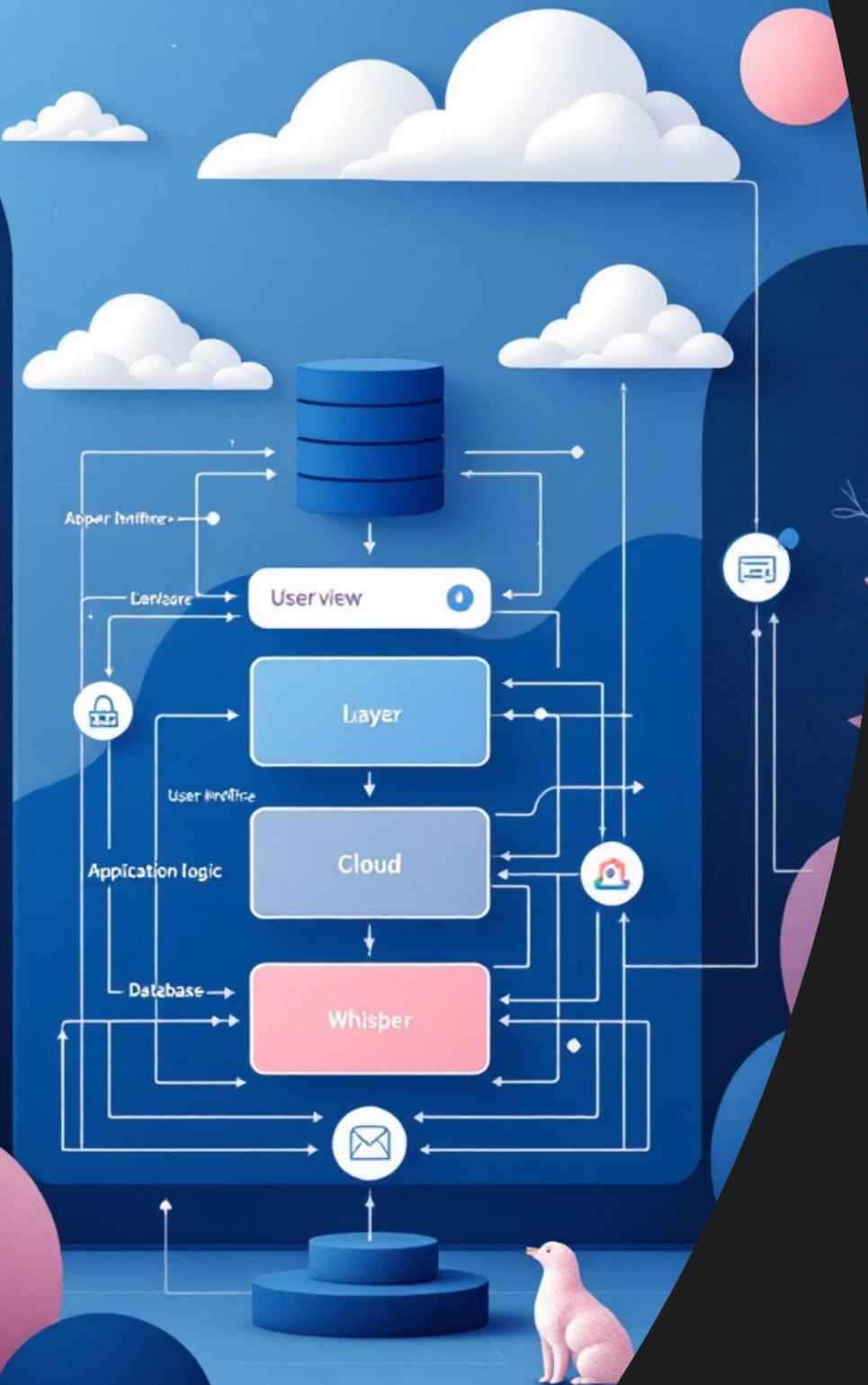
Backend Layer

Powered by Python and Flask, this layer orchestrates data processing, feature scaling, and communication between the interface and the model.

03

Model Layer

Houses the pre-trained Random Forest model saved as a .joblib file, enabling rapid predictions with consistent accuracy.



Data Science Workflow

01

Data Preprocessing

Handle missing values, remove outliers, and clean the dataset

02

Feature Scaling

StandardScaler normalizes features to mean=0, std=1

03

Model Training

Random Forest Regressor learns patterns from 80% training data

04

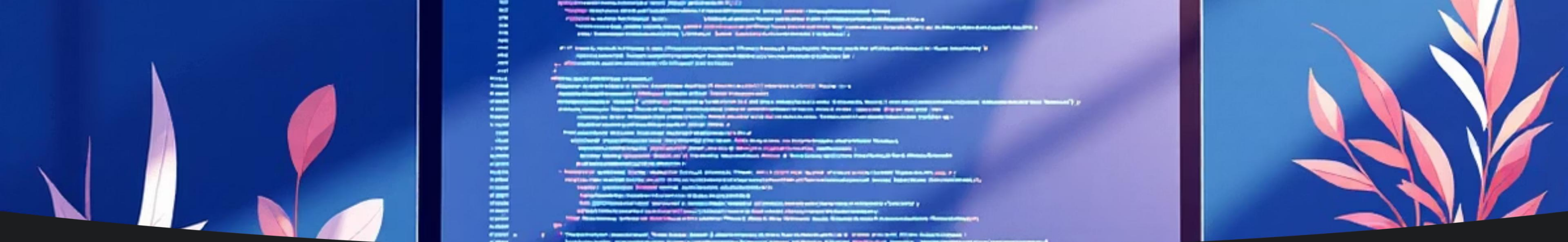
Evaluation

Test on 20% holdout set using MAE and R^2 metrics

05

Deployment

Serialize model and deploy via Flask API



Technology Stack



Frontend Technologies

HTML5, CSS3, and JavaScript create a responsive, user-friendly interface that works seamlessly across devices.



Backend Framework

Python with Flask provides a lightweight yet powerful foundation for API endpoints and business logic.



ML Libraries

Scikit-learn for model building [ML], Pandas for data manipulation and, and NumPy for numerical computations.



Development Environment

Visual Studio Code served as our integrated development environment for coding, testing, and debugging.



Key Input Features



Median Income

Neighborhood wealth indicator in 10k USD units



Property Details

Average rooms, bedrooms, and house age



Population

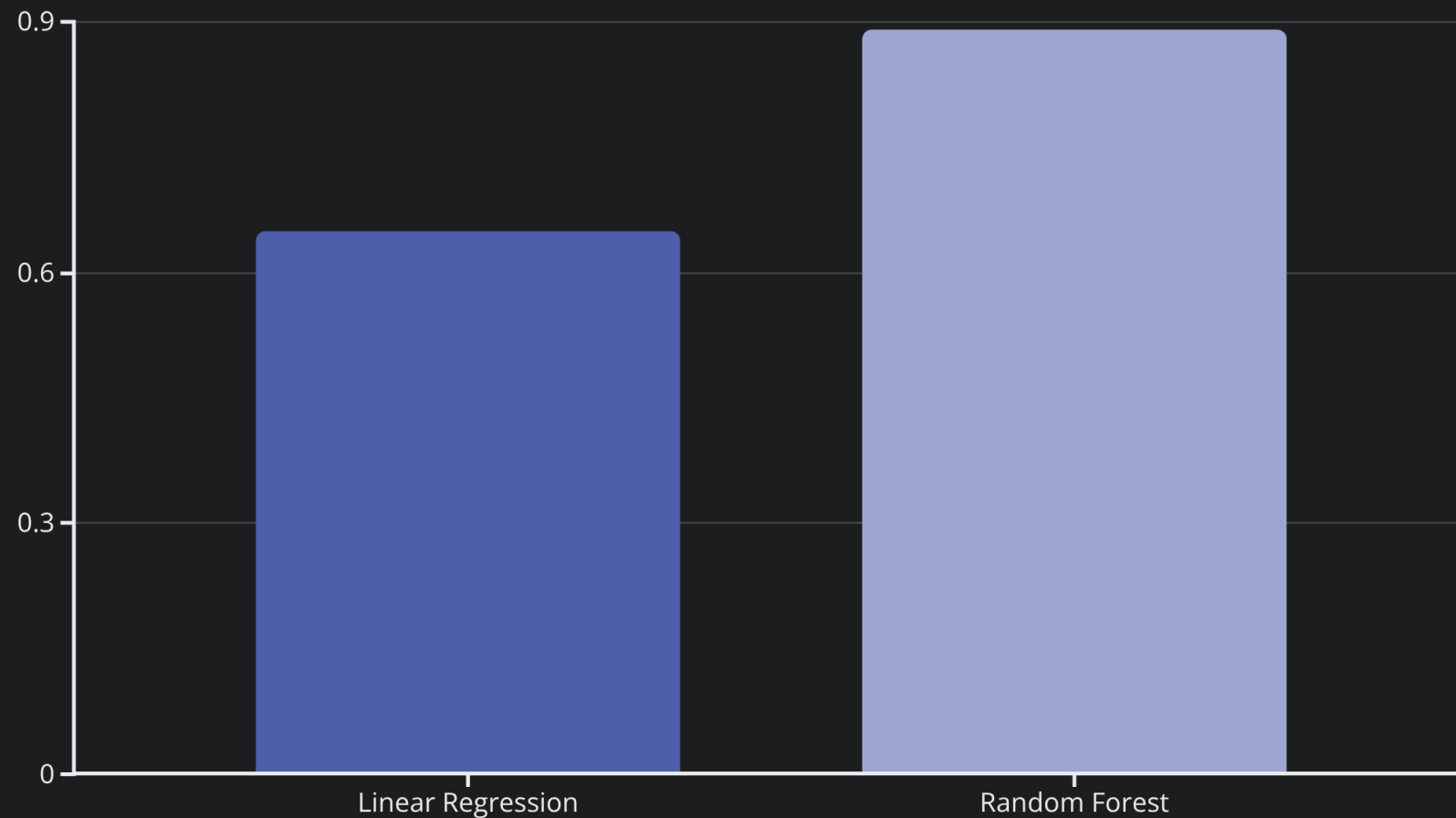
Area population density metrics



Geographic Data

Latitude and longitude coordinates

Model Performance



Superior Accuracy

Random Forest achieved R^2 score of 0.89, significantly outperforming Linear Regression baseline.

Key Predictors:

1. Median Income (strongest)
2. Latitude & Longitude
3. Average Rooms & House Age



Live Application Demo

Simple Input Process

Users enter essential property details through an intuitive form:

- Median household income
- Average number of rooms
- Property age and location
- Population and household metrics

Instant Results

The system processes data in real-time and displays:

- Estimated property value in USD
- Automatic conversion to INR
- Confidence metrics and key factors
- Visual breakdown of price drivers

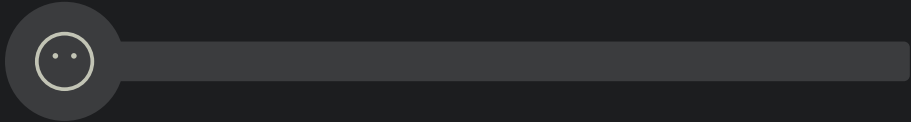
The dual-currency display makes the tool accessible to international investors and local buyers alike, eliminating the need for manual conversion.





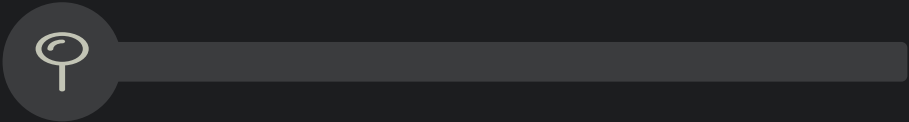
Future Roadmap

We envision several enhancements that will elevate the platform's capabilities and expand its market reach.



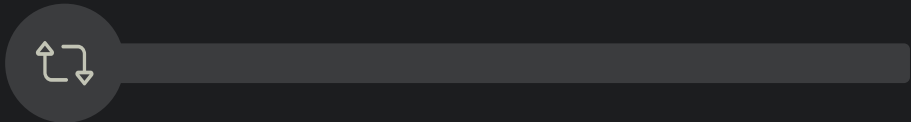
Deep Learning Integration

Implementing neural networks and advanced architectures to capture even more complex patterns and improve prediction accuracy beyond current benchmarks.



GIS Mapping Features

Incorporating Geographic Information Systems to analyse environmental factors like proximity to amenities, green spaces, transport hubs, and infrastructure quality.



Real-Time Market Data

Connecting to live property market APIs to provide dynamic valuations that reflect current trends, seasonal variations, and economic indicators.

Transforming Real Estate Valuation



Project Impact

The House Price Prediction System successfully replaces subjective manual valuation with a transparent, scientifically rigorous approach.

Key achievements:

- Eliminates human bias and inconsistency
- Delivers instant results at scale
- Reduces transaction time significantly
- Provides transparent, data-backed valuations

85%

Accuracy Rate

R^2 score demonstrates strong predictive power

10x

Faster Processing

Compared to traditional manual appraisals

100%

Objective Results

Zero human bias in price estimation